

Deflated Sharpe Ratio & Probability of Backtest Overfitting

A companion to *Convex Core and the DNSR Model Suite: A Deterministic ETF Allocation Framework for Drawdown-Controlled Compounding*

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Research system: the DNSR Agentic AI pipeline, directed and reviewed by the principal. No large-language model participates in the portfolio math.

Scope. This addendum is a standalone supplement. **The v1.0.2 paper is unchanged.** It is distributed with the same reproducibility companion (<https://github.com/DNSReeves/convex-core-paper>, concept DOI

10.5281/zenodo.20693931) and addresses the single piece of future work the review board left open: quantifying how much of Convex Core's risk-adjusted record could be a multiple-testing or backtest-overfitting artifact.

Summary. Treating the model's two continuous dials (`w_equity` × `vol_target` , 28 configurations) as the trial set over a 245-month panel (2006-02-28 → 2026-06-30), the deployed configuration's Sharpe survives multiple-testing deflation overwhelmingly: **Deflated Sharpe Ratio = 0.99998** (minimum track record ≈ 40 months vs. 245 available). The configurations are near-identical (annualized-Sharpe coefficient of variation 2.5%), and the **deployed configuration ranks 27th of 28** on Sharpe — it was chosen by a-priori risk posture, not by maximizing the backtest, so selection bias cannot have inflated it. The CSCV **Probability of Backtest Overfitting = 0.54** reflects that the dials are not meaningfully distinguishable in-sample (selecting a "best" dial is sampling noise), while every selected configuration remains profitable out-of-sample in 100% of 12,870 splits. The diagnostics corroborate the paper's parameter-minimal, deterministic design rather than revealing a tuned edge.

A1 · Why this test

A backtest's Sharpe ratio can be inflated two ways that have nothing to do with genuine edge: by **selection** (trying many configurations and reporting the luckiest) and by **non-normality** (fat tails and skew that make a given Sharpe less informative than the Gaussian assumption implies). The paper's defense is structural — the model is deterministic, its parameters fixed a-priori, and its negative results published — but the review board asked for the quantitative complement: the **Deflated Sharpe Ratio (DSR)**, which discounts the observed Sharpe for both the number of trials and the return distribution's shape, and the **Probability of Backtest Overfitting (PBO)**, which measures whether choosing the in-sample-best configuration generalizes out-of-sample.

A2 • Method

Trial set

The honest count of "trials" is the model's genuine search space. Convex Core exposes two continuous dials — the equity-weight target `w_equity` and the volatility target `vol_target`. We sweep `w_equity` $\in \{0.65, 0.70, 0.75, 0.80, 0.85, 0.90, 0.95\}$ $\times$ `vol_target` $\in \{0.10, 0.12, 0.14, 0.16\}$ = 28 configurations, each run through the *same* deterministic engine (same regime layer, sleeves, brake, and costs) and reduced to a monthly total-return stream. The deployed paper configuration is (`w_equity=0.95`, `vol_target=0.12`). All 28 streams share a common 245-month window.

Deflated Sharpe Ratio

The DSR is the probability that the deployed configuration's *true* Sharpe exceeds the *expected maximum Sharpe under the null* that no configuration has skill — where that expected maximum grows with the number of trials and the dispersion of Sharpe across them — evaluated through the Probabilistic Sharpe Ratio, which corrects for the sample's skewness and kurtosis and length. DSR > 0.95 means the result survives the multiple-testing penalty. (Bailey & López de Prado, 2014.)

Probability of Backtest Overfitting (CSCV)

Combinatorially-Symmetric Cross-Validation splits the 245-month panel into 16 contiguous blocks of 15 months. For every one of the 12,870 symmetric ways to assign half the blocks in-sample and half out-of-sample, it selects the configuration with the best in-sample Sharpe and records that configuration's out-of-sample rank. **PBO** is the fraction of splits in which the in-sample winner lands below the out-of-sample median. (Bailey, Borwein, López de Prado & Zhu, 2017.)

A3 • Result — parameter insensitivity and the Deflated Sharpe Ratio

Across the entire 28-configuration grid the annualized Sharpe spans only 1.024–1.115 (mean 1.075, standard deviation 0.027, coefficient of variation **2.5%**). The model is strikingly insensitive to its dials: there is no poor configuration to avoid and no standout configuration to cherry-pick. Tellingly, the **deployed configuration ranks 27th of 28** — near the bottom on Sharpe — because `w_equity=0.95` was selected for accumulation-phase upside capture (an a-priori risk-posture choice), not to flatter the backtest.

Quantity	Value
Deployed configuration	w_equity 0.95, vol_target 0.12
Observations (months)	245
Number of trials (N)	28
Sharpe — monthly (annualized)	0.299 (1.03)
Skewness	-0.35
Kurtosis (normal = 3)	2.90
Variance of Sharpe across trials	6.29e-05
Expected max Sharpe under null — annualized	0.056
Probabilistic Sharpe vs. 0	0.99999
Deflated Sharpe Ratio	0.99998
Min. track record for 95% significance	40 months

Table 1. Deflated Sharpe Ratio for the deployed configuration. The multiplicity penalty is small because the trials barely disperse; the near-normal shape (kurtosis 2.90, mild negative skew) leaves the Sharpe essentially intact. DSR ≈ 1 with a 40-month minimum track record against 245 months observed.

A4 • Result — Probability of Backtest Overfitting

Quantity	Value
Method	CSCV (16 blocks $\times$ 15 months, 12,870 splits)
Probability of Backtest Overfitting	0.537
Median OOS rank of the in-sample best (0.5 = random)	0.46
P(selected configuration's OOS Sharpe < 0)	0.000
OOS-vs-IS degradation slope	-0.96

Table 2. CSCV overfitting diagnostics over 240 months.

PBO = 0.54 reads, against a naïve <0.5 rule of thumb, as a borderline figure — but the correct reading is that it is the **expected null** here. CSCV asks whether the in-sample-best configuration generalizes; when the configurations are near-identical (Sharpe CoV 2.5%, Table 1), there is no genuinely-best configuration to identify, so the in-sample winner is chosen by sampling noise and lands near the out-of-sample median by construction (median OOS rank 0.46). The diagnostic that actually carries signal is the out-of-sample outcome of the selected configuration: it is **profitable in 100% of the 12,870 splits** ($P(\text{OOS Sharpe} < 0) = 0.000$). The mildly negative degradation slope (-0.96) is precisely why the model fixes its dials a-priori instead of optimizing them — chasing the highest in-sample Sharpe hands a

little back out-of-sample. None of this is overfitting in the sense PBO was designed to expose (a tuned strategy that collapses out-of-sample); it is the fingerprint of a parameter-insensitive design.

A5 • Interpretation — what this does and does not establish

It establishes: (1) the deployed Sharpe is not a multiple-testing illusion — it survives deflation with a 40-month minimum track record against 245 months of data; (2) the model is parameter-insensitive over its dial space (2.5% Sharpe CoV), so there is no configuration-selection edge to overfit; (3) the deployment choice was not Sharpe-maximizing (rank 27/28), foreclosing the cherry-picking failure mode by construction; (4) across 12,870 out-of-sample folds the family never loses money.

It does not establish live performance, edge over the strongest balanced benchmark on risk-adjusted ratios (the paper's §8 is explicit that the advantage over 40/60 is in return, not risk), or robustness to *structural* choices outside the two dials swept here (sleeve definitions, regime cutoffs, universe) — those are probed by the §13 ablation grid and §6 walk-forward, not by this test. CSCV's PBO is also least informative exactly when trials are near-identical, as they are here; we therefore lean on the accompanying out-of-sample loss probability and the non-optimized deployment rank rather than the PBO point estimate alone. These results sharpen, and are consistent with, the paper's §3.1 (determinism, no fitted parameters) and §15 (limitations).

A6 • Reproducibility

Computed by `scripts/research/pub_tier2/run_dsr_pbo.py` from the same deterministic engine, warehouse, and monthly basis as the paper; the artifact is `results/dsr_pbo.json`. Re-run from the companion repository:

```
python scripts/research/pub_tier2/run_dsr_pbo.py    # writes results/dsr_pbo.json
python report/build_addendum.py                    # renders this addendum (HTML + PDF)
```

References — Bailey, D. & López de Prado, M. (2014), *The Deflated Sharpe Ratio*, Journal of Portfolio Management 40(5). · Bailey, D., Borwein, J., López de Prado, M. & Zhu, Q. (2017), *The Probability of Backtest Overfitting*, Journal of Computational Finance 20(4).

Disclaimer. This addendum presents hypothetical, backtested model performance for research and decision-support purposes only. The results were not achieved by an actual client account and do not reflect the impact of individual investor circumstances, taxes, custody fees, advisory fees, account restrictions, or implementation delays. Backtested performance is inherently limited because it is constructed with the benefit of historical data and may not reflect conditions that would have existed during live trading. Past performance does not guarantee future results. This is not investment advice, an offer, or a solicitation. Research conducted by the

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